

IBM ECM System Monitor Field Guides

Integration with 3rd party tools
(Email, IT Service Management and Message Channels)

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Introduction

Overview

This guide describes in detail how IBM ECM System Monitor (ESM) can be integrated with 3rd party tools.

Disclaimer

The content of this document is based on ESM Version 5.7.0.0-000. The descriptions and guidelines in this document are for informational purposes only. Up-to-dateness, content completeness, appropriateness and validity for all possible scenarios cannot be guaranteed. All information is provided on an as-is basis. The author is not liable for any errors or omissions in this document or any losses, injuries and damages arising from its use.

If you are planning to set up or configure ESM or to adjust an existing installation, it is absolutely necessary to consider current security whitepapers, release notes and announcements from the official IBM ECM System Monitor product documentation website.

Purpose of this guide

This guide provides a step-by-step setup for integration of ESM with 3rd party tools using tasks.

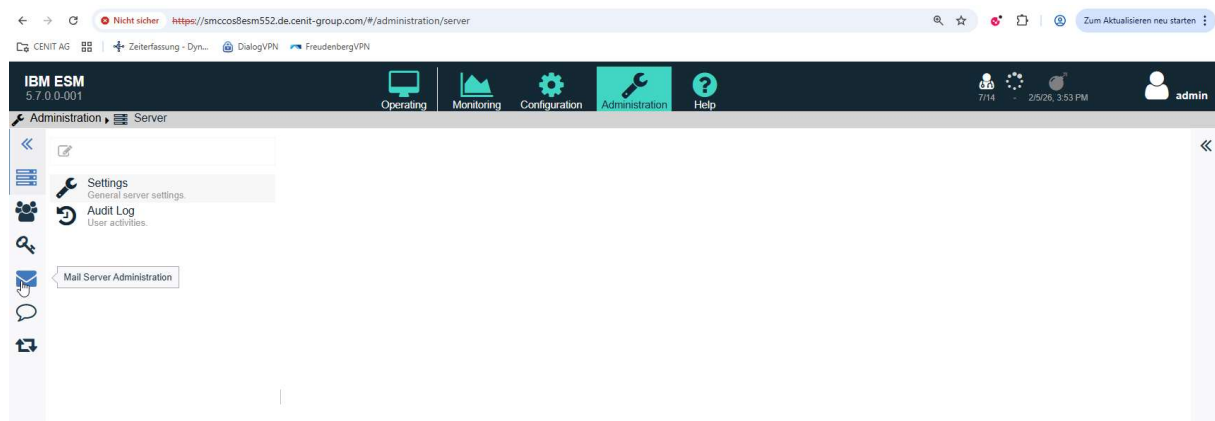
Requirements

- ESM Version 5.7.0.0-000 or later (already installed).

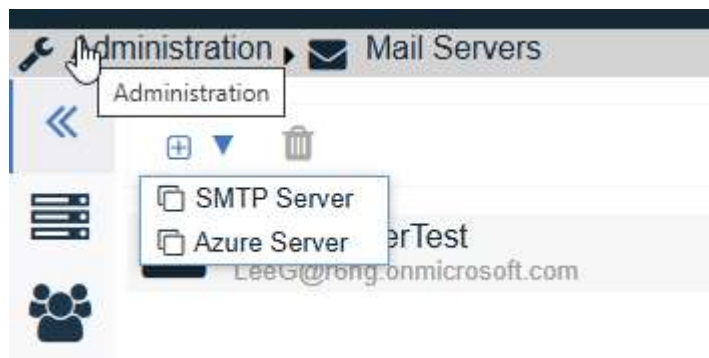
1. Integrating ESM with SMTP (Mail)

1.1 Prepare the mail server configuration in ESM

ESM offers two mail server options – either a standard smtp server or an Azure server. For both options you need to switch into the Configuration console and select Administration dashboard from the top. In the menu on the left, you find an entry for the mail server administration which you need to select.



After selecting it, you can click on the drop-down next to the +. In the drop-down you will find the two options. Select the one that works for you.



Example for SMTP Server:



*Server Name:	MySmtServerAlias
*Host:	SMTPServerHostname
*Port:	25
*From Address:	MySenderAdress@MyDomain.com
Authentication:	☒
* Username:	MyUserNamemyDomain
* Password:	*****
▶ Additional Parameters	

The Server Name is just an alias, like a configuration name. You can specify whatever you want here. For the other parameters please specify your relevant information.

Authentication might be required but, in some cases, it is not necessary.

There is also an option for additional parameters. They are usually not required.

Save your setup by clicking on the check icon above the configuration.

Example for Azure Server:

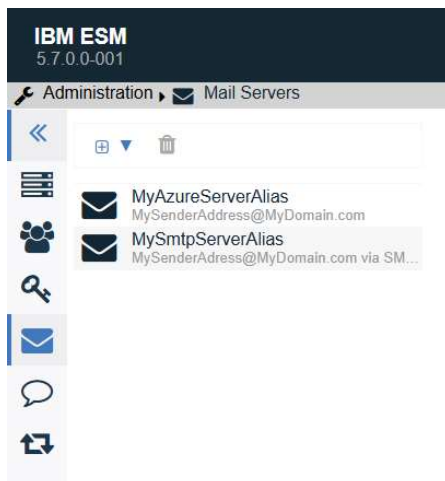
 	
*Server Name:	MyAzureServerAlias
*From Address:	MySenderAddress@MyDomain.com
*Tenant ID:	532ef83b-73ff-4909-abfc-2e161f50253d
*Client ID:	d7cbdd40-4579-4320-a5ed-b6aa397fa1bd
Secret:	*****
Private Key Path:	
Certificate Path:	

The Server Name is just an alias, like a configuration name. You can specify whatever you want here. For the other parameters please specify your relevant information.

Private Key Path and Certificate Path might be required but usually this is not necessary.

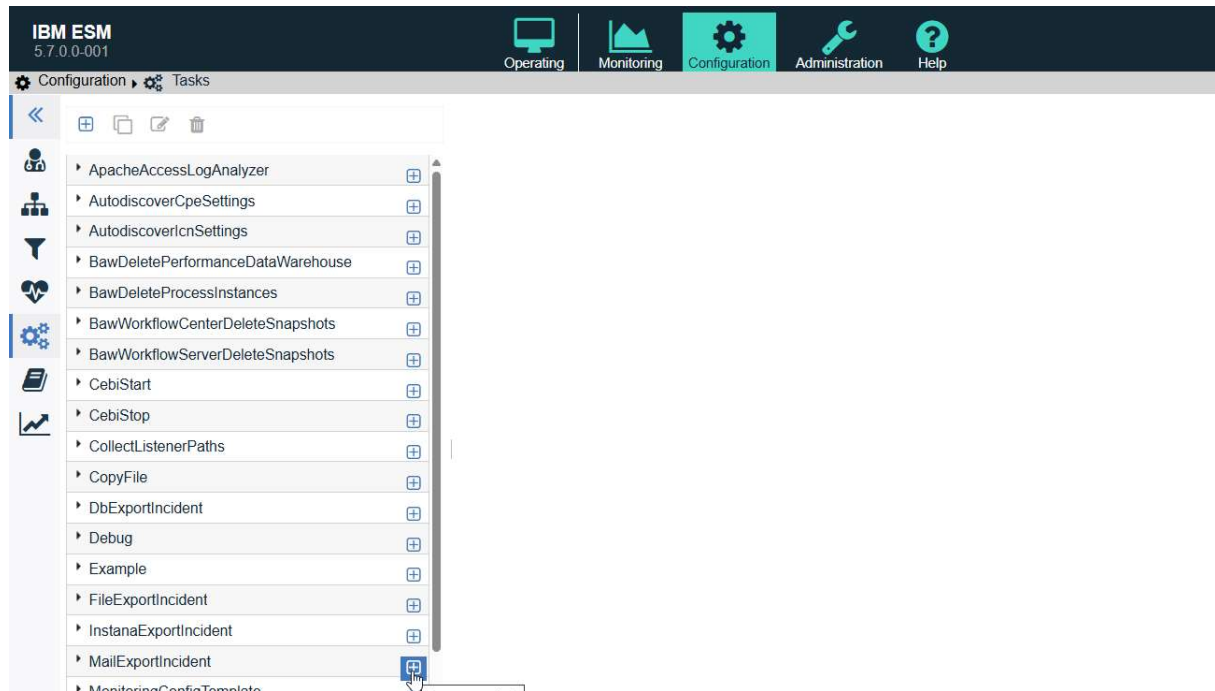
Save your setup by clicking on the check icon above the configuration.

After saving you should see your mail server on the list.



1.2 Preparing the MailExportIncident task

Open the Configuration console and switch to the configuration dashboard. Select Task in the menu on the left and click on the + next to MailExportIncident to create a new task:



The configuration for the task will open. Please adjust the setup as required.

✕ < > ✓ ↻

⚙️ Task Configuration

Select the type of the task and enter its configuration data.

*Name

ForwardIncidentByMail

active ☐

Task:

MailExportIncident

Subject Template: \$\$Situation.name - \$Severity

Body Template: \$Timestamp: \$Severity: \$Value | \$Message | \$Error

Mail Server Name: MySmtServerAlias

Recipients: YourRecieipient1@YourDoamin.com,YourRecieipient2@YourDoamin.com

👤 Agent

Select the location where the Task should run.

[Run on Server]

📅 Schedule

If active when and how often should the Task be executed?

☒ Interval ☐ Cron

Start:

End:

Interval:

10

Minute(s)

Specify a name for the task – make sure you will be able to easily identify it later.

Keep “active” unchecked, as the task will be triggered by criterias of the incident.

You can adjust Subject Template and Body Template – see the configuration help that is offered on the right side of the setup.

For Mail Server Name select your mail server that should be used from the drop-down.

In Recipients you can specify one recipient or a comma separated list of recipients.

As Agent please specify [Run on Server].

The Schedule is only relevant if a task is set to active.

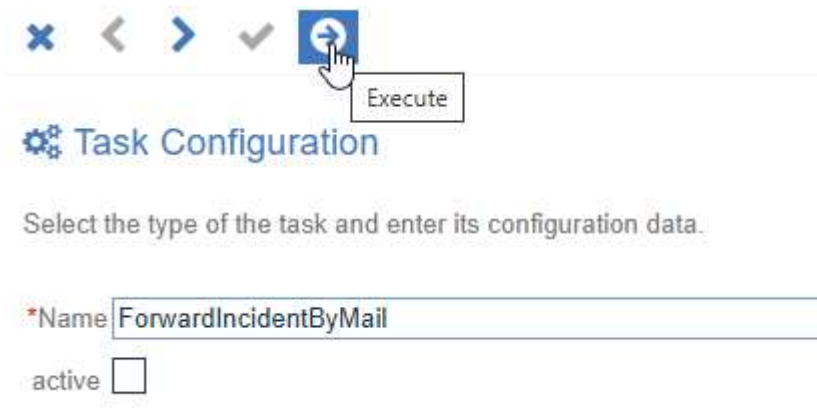
Save your setup by clicking on the check icon above the configuration – the configuration editor will close.

1.3 Test of the Task

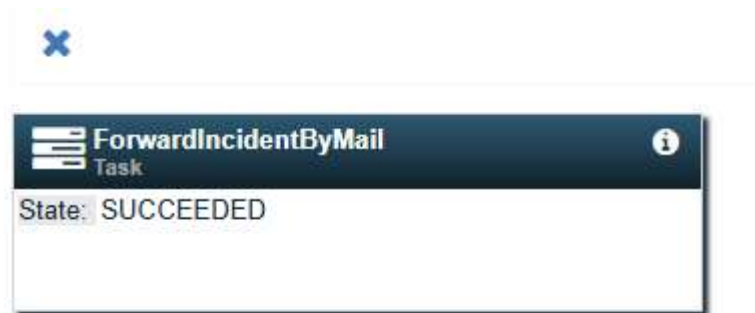
Open the task configuration again by double clicking on it. You find your task below the MailExportIncident task in the sidebar.



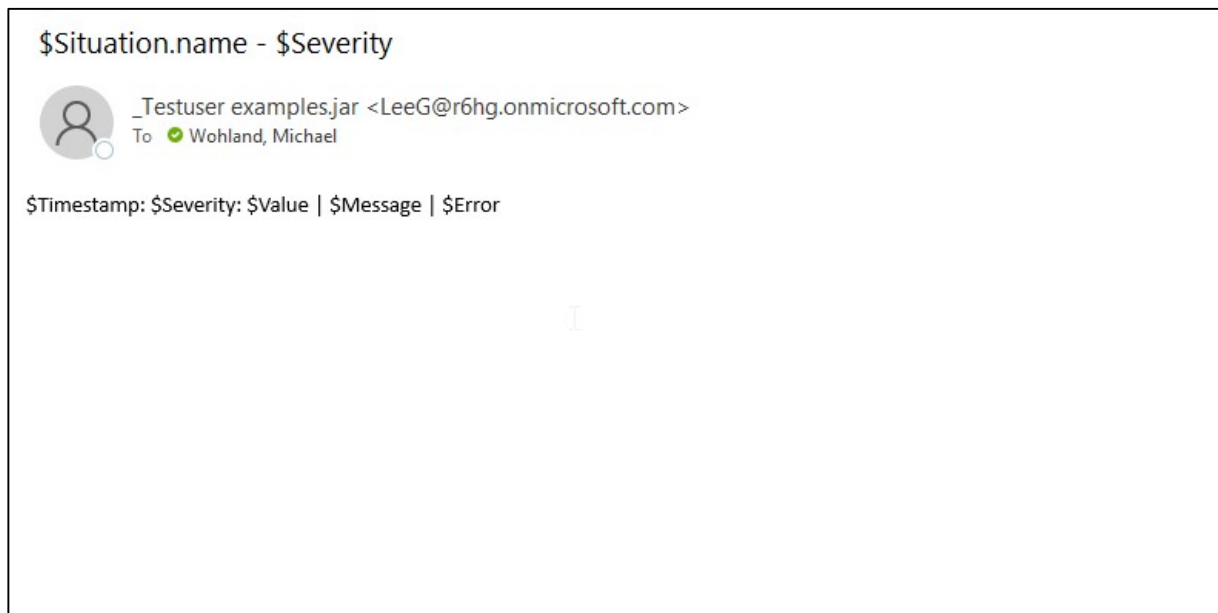
You can now execute the task manually as a test:



Now you should see a task result in the console:



If you have specified yourself in the recipients, you should also receive an email that looks like this – as this was a test execution the content will just show the template information:



You can now setup the task as automation (see chapter 4).

2. Integration with ITSM (example based on Service Now)

2.1 Find the description of the REST API and create the message template

Check for the description of the API on the vendor pages. E.g. this is the URL of the Web Service API from “ServiceNow”:

<https://docs.servicenow.com/bundle/quebec-it-operations-management/page/product/event-management/task/send-events-via-web-service.html>

On the “ServiceNow” page you can find this example for the JSON object with one record at a time:

```
{
  "records":
  [
    {
      "source": "SCOM",
      "event_class": "SCOM 2007 on scom.server.com",
      "resource": "C:",
      "node": "name.of.node.com",
      "metric_name": "Percentage Logical Disk Free Space",
      "type": "Disk space",
      "severity": "4",
      "description": "The disk C: on computer V-W2K8-dfg.dfg.com is running out of disk space.
The value that exceeded the threshold is 41% free space.",
      "additional_info": {
        "scom-severity": "Medium",
        "metric-value": "41",
        "os_type": "Windows.Server.2008"
      }
    }
  ]
}
```

The default template in the WebHookExportIncident task looks like this:

```
$Timestamp: $Severity: $Value | $Message | {"text" : "$Timestamp: $Severity: $Value |
$Message | $Error "} +
```

Incident internal fields are used with the following notation: \$EntityType.FieldInCamelCase. They can be seen when double clicking on an incident. The field must be defined in CamelCase, meaning each word starts with a capital letter. E.g. SituationCfFld. Some of the fields, e.g. Incident.ID, might not be available at task execution since the entry is created when added to the DB.

The following EntityTypes are available:

- Incident (default - used if nothing but FieldInCamelCase is specified)
- Situation
- Sample
- ProbeConfig
- Agent
- Subsystem

The information must be combined with the required JSON object in the actual template. This is just an example and may not fit in your environment. Some JSON object variables have been removed as they are not needed:

```
{
  "records":
  [
    {
      "source":"$Source",
      "event_class":"$Classification",
      "node":"$Agent.HostName",
      "metric_name":"$Probe.Name",
      "severity":"4",
      "description":"$Message",
      "additional_info":{"This alert pertains to XYZ System - $Timestamp | $Error"
                        }
    }
  ]
}
```

The template is completed, make sure to put it in a one line format after you have created it. We recommend to do that in an editor like notepad++ or vim.

```
{"records": [ { "source":"$Source", "event_class":"$Classification",
"node":"$Agent.HostName", "metric_name":"$Probe.Name", "severity":"4",
"description":"$Message", "additional_info":{"This alert pertains to XYZ System -
$Timestamp | $Error" } } ] }
```

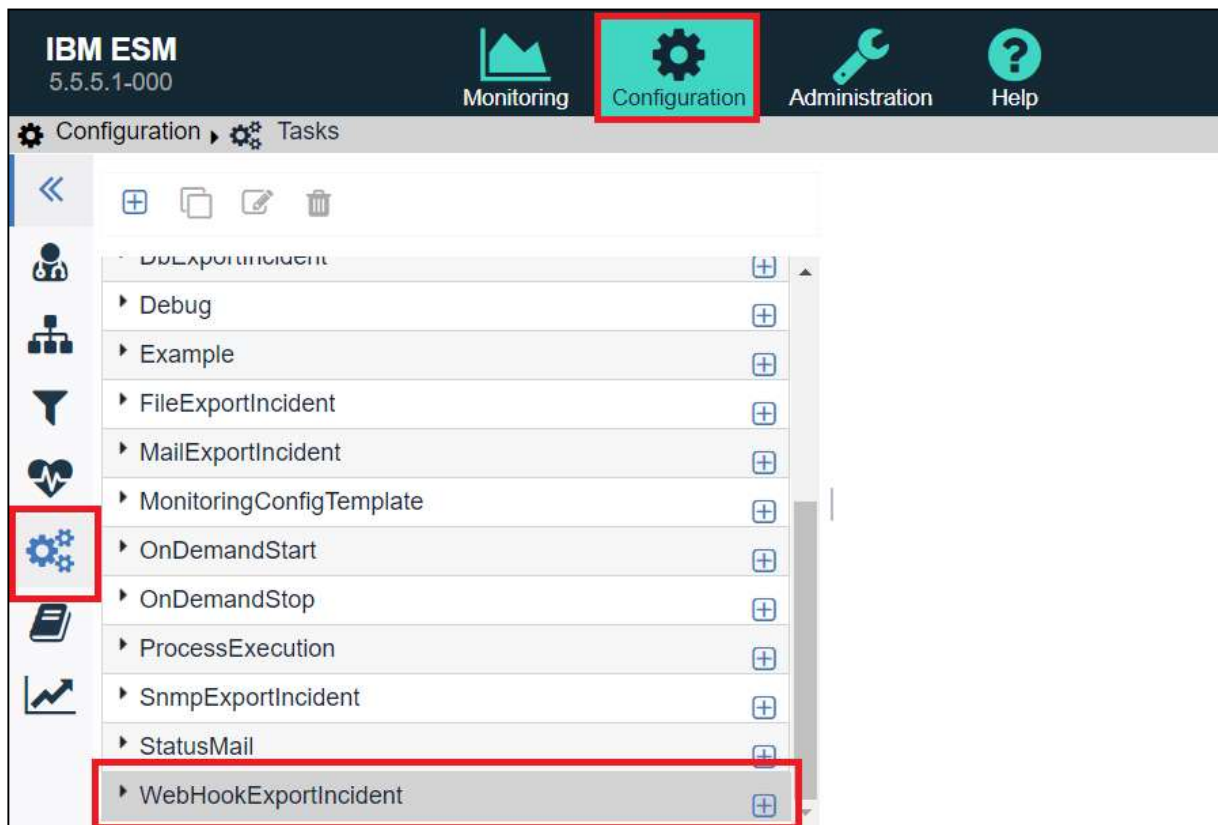
The severity could not be replaced in this case by using the \$Severity info from the incident, as the information would be something like HARMLESS or CRITICAL. The object needs a number in this case. So depending on how many different severities for objects are needed in the Service Management Tool, you might end up using several ESM tasks (one for each severity).

2.2 Setup of the WebHookExportIncident task

Once the template is created, you provide the URL where the JSON object will be sent to. In case of “ServiceNow” an example is given on the above specified web page again. As you can see there are various URLs for single or multiple records that are provided in the JSON object. Because the task will be triggered by one incident only, the JSON object will only contain one record. Therefore the URL is specified with this default like on the web page:

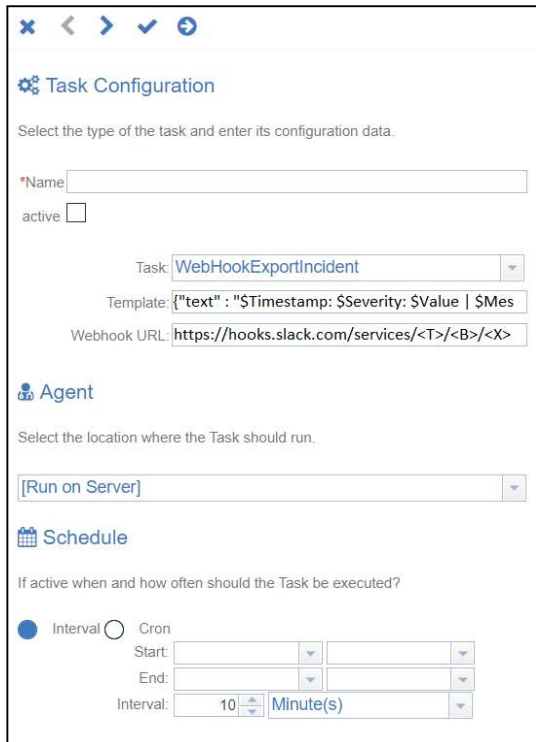
<http://<instancename>.service-now.com/api/global/em/jsonv2>

Now the actual task setup is done in the ESM console. Browse to the configuration and on the left to the tasks. There you will find the WebHookExportIncident task.

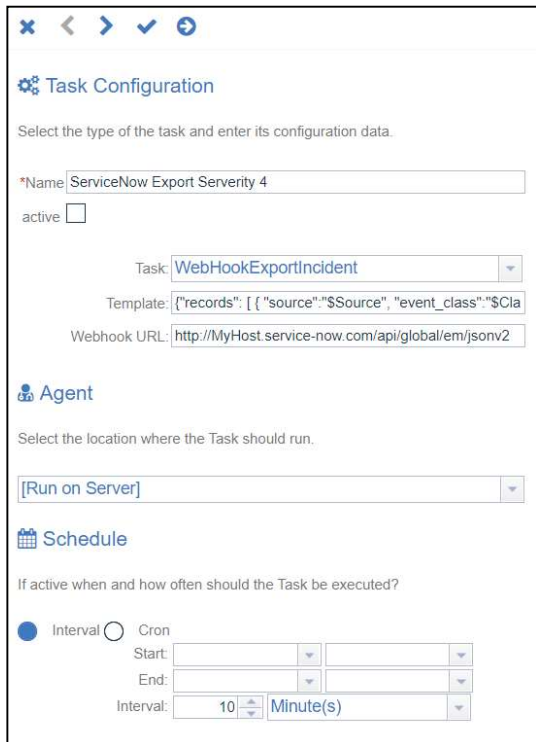


Click on the + on the right of the WebHookExportIncident task to open the task editor.

The editor looks like this:



The screenshot shows the 'Task Configuration' editor. At the top, there are navigation icons (back, forward, cancel, save, refresh). The title is 'Task Configuration' with a gear icon. Below the title is the instruction 'Select the type of the task and enter its configuration data.' The form includes: a text field for '*Name' (empty), an 'active' checkbox (unchecked), a 'Task' dropdown menu (set to 'WebHookExportIncident'), a 'Template' text field (containing '{ "text" : "\$Timestamp: \$Severity: \$Value | \$Mes'), and a 'Webhook URL' text field (containing 'https://hooks.slack.com/services/<T>//<X>'). Below this is the 'Agent' section with the instruction 'Select the location where the Task should run.' and a dropdown menu (set to '[Run on Server]'). The 'Schedule' section has the instruction 'If active when and how often should the Task be executed?' and two radio buttons: 'Interval' (selected) and 'Cron' (unselected). The 'Interval' section has 'Start' and 'End' time pickers (both empty) and an 'Interval' field (set to '10 Minute(s)').



The screenshot shows the 'Task Configuration' editor with customized settings. The title is 'Task Configuration' with a gear icon. Below the title is the instruction 'Select the type of the task and enter its configuration data.' The form includes: a text field for '*Name' (containing 'ServiceNow Export Severity 4'), an 'active' checkbox (unchecked), a 'Task' dropdown menu (set to 'WebHookExportIncident'), a 'Template' text field (containing '{ "records": [{ "source": "\$Source", "event_class": "\$Cla'), and a 'Webhook URL' text field (containing 'http://MyHost.service-now.com/api/global/em/jsonv2'). Below this is the 'Agent' section with the instruction 'Select the location where the Task should run.' and a dropdown menu (set to '[Run on Server]'). The 'Schedule' section has the instruction 'If active when and how often should the Task be executed?' and two radio buttons: 'Interval' (selected) and 'Cron' (unselected). The 'Interval' section has 'Start' and 'End' time pickers (both empty) and an 'Interval' field (set to '10 Minute(s)').

- Specify a name for the task, e.g. ServiceNow Export Severity 4.
- Do not check the active button as the task will be triggered by an incident.

- Copy the template that has been created in the template section.
- Copy the matching URL in the Webhook URL.
- As Agent use [Run On Server] → The task will be executed on the actual ESM Server which is what we want.
- Because the task is triggered by an incident, the schedule part will not be used at all.
- Save the task by clicking on the hook button above the “Task Configuration”.

You should now see the saved task specified as “DEACTIVATED” below the WebHookExportIncident entry in the task list.



Repeat this step for tasks that use other severities, e.g. three more tasks for severities 1-3.

You can now setup the task as automation (see chapter 4).

3. Integration with message channel

3.1 Prepare the message channel to accept webhooks

Example for MS Teams:

For MS Teams there is a webpage available that describes how an incoming webhook connector can be set up for channels. Their URL is:

<https://learn.microsoft.com/en-us/microsoftteams/platform/webhooks-and-connectors/how-to/add-incoming-webhook?tabs=dotnet>

At some point of the setup you will receive a webhook URL, which you need for the integration with ESM. Note it or copy it to a document. The URL looks like this:

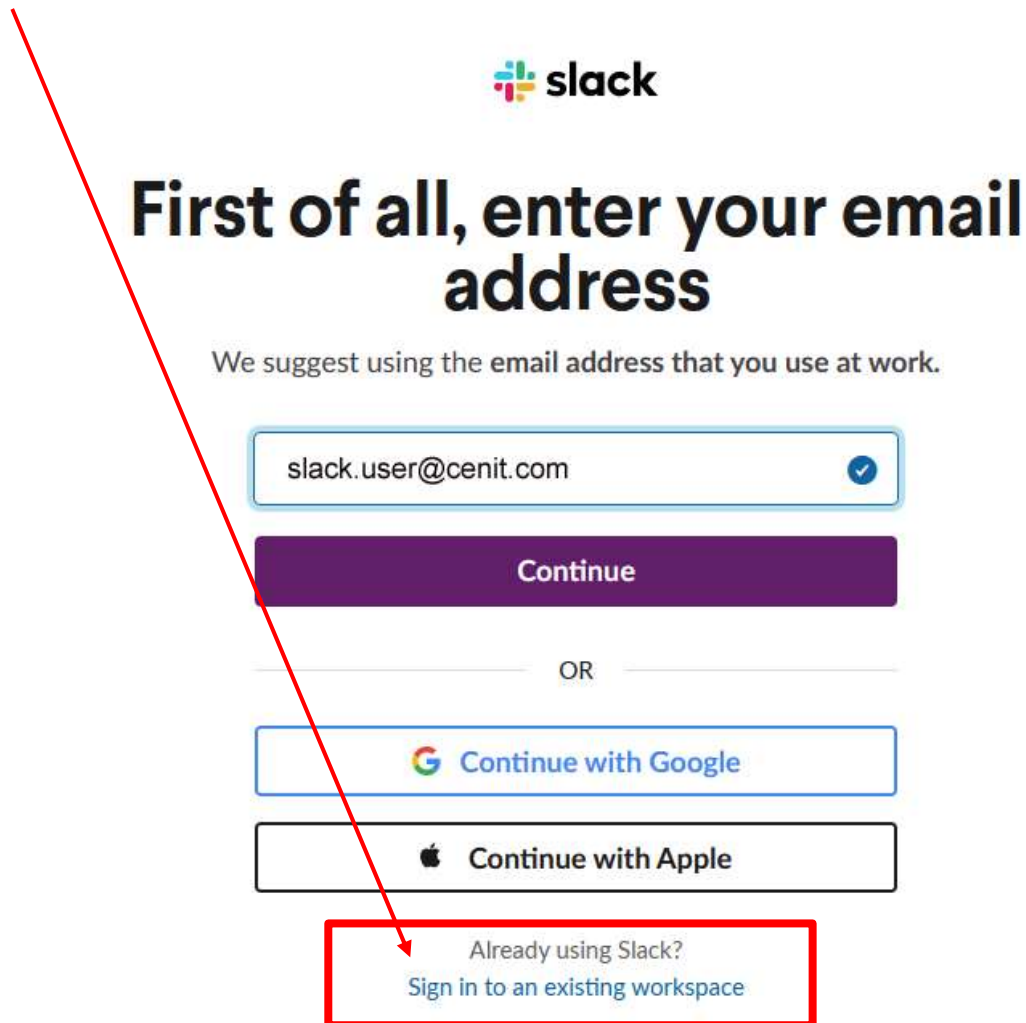
<https://cenitgroupglobal.webhook.office.com/webhookb2/baa589c7-5f5b-4a9d-89d3-7a4cbab7b71b@ce368477-a736-4353-b4ef-741ef2740d36/IncomingWebhook/343157e99ad946e5a166dd1d8ab5c8e7/a6858fdd-a65f-4093-867b-45a5958a51f1>

Example for Slack:

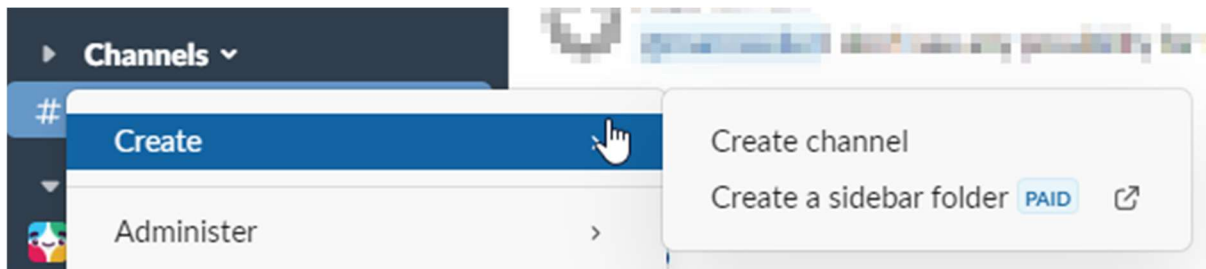
For creating a workspace in Slack there is a “how to” available.

The page URL is: <https://slack.com/help/articles/206845317-Create-a-Slack-workspace>

Login to Slack and open your workspace – in this case it is **ESM-CSM-Workspace**.

The image shows the Slack login interface. At the top is the Slack logo. Below it is the heading "First of all, enter your email address" in large, bold, black text. Underneath the heading is a subtext: "We suggest using the email address that you use at work." There is a text input field containing the email address "slack.user@cenit.com" with a blue checkmark icon on the right. Below the input field is a purple "Continue" button. Underneath the button is a horizontal line with the word "OR" in the center. Below the line are two buttons: "Continue with Google" (with the Google logo) and "Continue with Apple" (with the Apple logo). At the bottom, there is a red rectangular box containing the text "Already using Slack?" and "Sign in to an existing workspace" in blue. A red arrow points from the top left of the page towards the "Sign in to an existing workspace" link.

Once you have signed in, you can create a new channel or use an already existing one. For a new one, select the drop down at Channels and specify Create -> Create Channel:



Specify a name and create:

create channels ×

Channels are used by your team to communicate. Ideally, they should be topic-related - #marketing, for example.

 This name is already used for a channel, user or user group

Name 64

esm-webhook-test

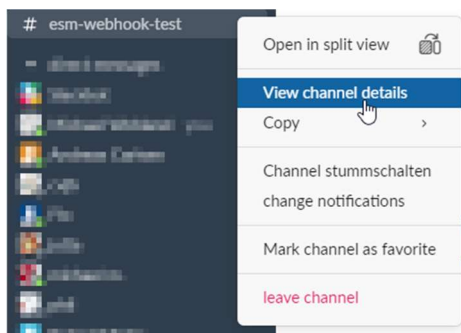
Description (optional)

What is this channel about?

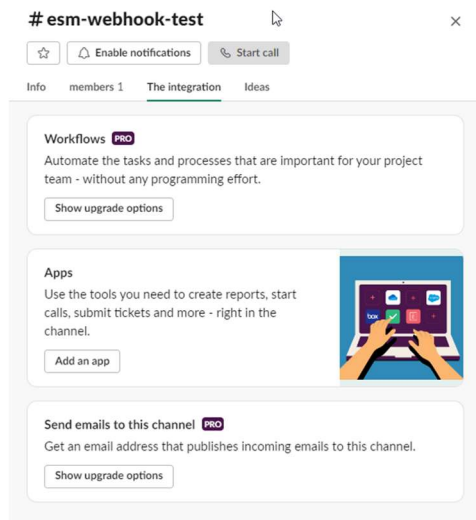
Set to "Closed".
When a channel is converted to a private channel, viewing and joining is invitation-only. ☐

Create

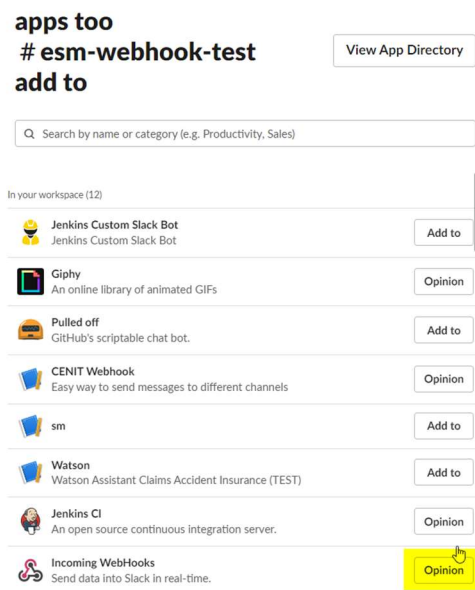
Right click on the channel and specify "View channel details":



In the next window that opens, select the integration tab and click on “Add an app”:

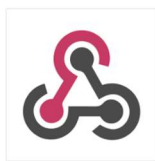


In the next window that opens, select “Opinion” next to Incoming Webhook:



Then select configuration:

Incoming webhooks



Incoming webhooks

Please note, this is a legacy custom integration - an outdated way for teams to integrate with Slack. These integrations lack newer features and they will be deprecated and possibly removed in the future. **We do not recommend their use.** Instead, we suggest that you check out their replacement: [Slack apps](#).

Incoming Webhooks are a simple way to post messages from external sources into Slack. They make use of normal HTTP requests with a JSON payload, which includes the message and a few other optional details described later.

[Message Attachments](#) can also be used in Incoming Webhooks to display richly-formatted messages that stand out from regular chat messages.

configuration

Add to Slack:

< Browse apps



Add to Slack

More info and support

[Get app support](#)

[Privacy Policy](#)

[Conditions](#)

Incoming webhooks

Description

configuration

Security & Compliance

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Select the channel that was just created or that you want to use for forwarding the info:



Incoming webhooks

Send data into Slack in real time.

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**New to Slack integrations?**

Check out our [getting started](#) guide to familiarize yourself with the most common integrations. You will also get a lot of tips that you should keep in mind when developing. You can also [register as a developer](#) to let us know what you're working on and to receive future updates on our APIs.

Post to channel

Start by choosing a channel for your incoming webhook to post messages to.

Wähle einen Channel...

Choose a channel...

- Michael Wohland Michael Wohland
- # generally
- Andreas Carlsen Andreas Carlsen
- cvjb Christian Brussow
- # esm webhook test
- Flo Flo
- # Freudenberg

Click on “Add inbound webhooks integration”:

Post to channel

Start by choosing a channel for your incoming webhook to post messages to.

esm webhook test

or create new channel

Add inbound webhooks integration

When you create an incoming webhook, you agree to the [Slack API Terms of Service](#).

The final page shows the Webhook URL along with other useful information such as possible message designs – Note down the Webhook URL:

Browse apps > Custom integrations > Incoming webhooks > Edit configuration

 **Your workspace has reached the maximum number of integrations.** [Upgrade your plan](#) to get unlimited integrations.

 **Incoming webhooks**
Added by Michael Wohland on March 21, 2023 Deactivate • Removed

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Setup Guide

We'll walk you through all the necessary steps to configure an incoming webhook so you can quickly start sending data to Slack. close

Webhook URL `https://hooks.slack.com/services/TFDJEV4HL/B04UJ4K488Z/mCcP6hoa1UgJg45GEmmLd1f`

send messages You have two options for sending data to the above webhook URL:

3.2 Preparing the JSON for forwarding the ESM information

Example for MS Teams:

For MS Teams it is possible to define a variety of JSON strings with different designs for the information that is forwarded. In this example here I am using the so called "Message Card" format. Here you can find more information for the creation of such cards:

<https://learn.microsoft.com/en-us/microsoftteams/platform/webhooks-and-connectors/how-to/connectors-using?tabs=cURL>

Message Card example for MS Teams:

```
{
  "@type": "MessageCard",
  "@context": "http://schema.org/extensions",
  "title": "ESM - Incident created",
  "themeColor": "0076D7",
  "summary": "ESM Webhook Test",
  "sections": [{
    "title": "$Situation.name - $Severity",
    "facts": [{
      "name": "Timestamp",
      "value": "$Sample.Timestamp"
    }, {
      "name": "Message",
      "value": "$Message"
    }, {
      "name": "Error:",
      "value": "$Error"
    }, {
      "name": "Value:",
      "value": "$Value"
    }
  ]],
  "startGroup": true,
  "markdown": true
},
{
  "potentialAction": [{
    "@type": "OpenUri",
    "name": "Connect to ESM Server",
    "targets": [{
      "os": "default",
      "uri": "https://myESMServer:9443/"
    }
  ]
}
]
```


As you can see in the facts section, there are name/value pairs. The name will be shown as identifier and the value is the related information from ESM. The \$Something is replaced with the actual information from the incident. This section can be enhanced as you need it.

Create a single line out of the above for later usage:

```
{ "@type": "MessageCard", "@context": "http://schema.org/extensions", "title": "ESM - Incident created", "themeColor": "0076D7", "summary": "ESM Webhook Test", "sections": [{ "title": "$Situation.name - $Severity", "facts": [{ "name": "Timestamp", "value": "$Sample.Timestamp" }, { "name": "Message", "value": "$Message" }, { "name": "Error:", "value": "$Error" }, { "name": "Value:", "value": "$Value" }], "startGroup": true, "markdown": true }, { "potentialAction": [{ "@type": "OpenUri", "name": "Connect to ESM Server", "targets": [{ "os": "default", "uri": "https://myESMServer:9443/" } ] } ] }
```

Example for Slack:

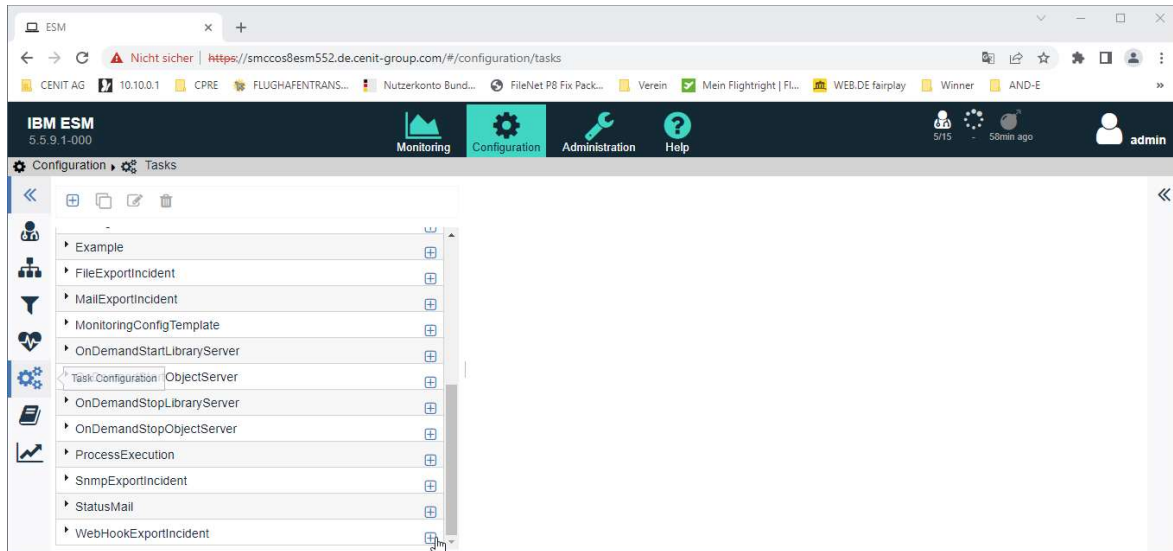
Single line example based on the message design from the final page when creating the Webhook URL above:

```
{ "text": "myservername | $Sample.Timestamp| $Severity| $Value| $Message| $Error| Classification" }
```

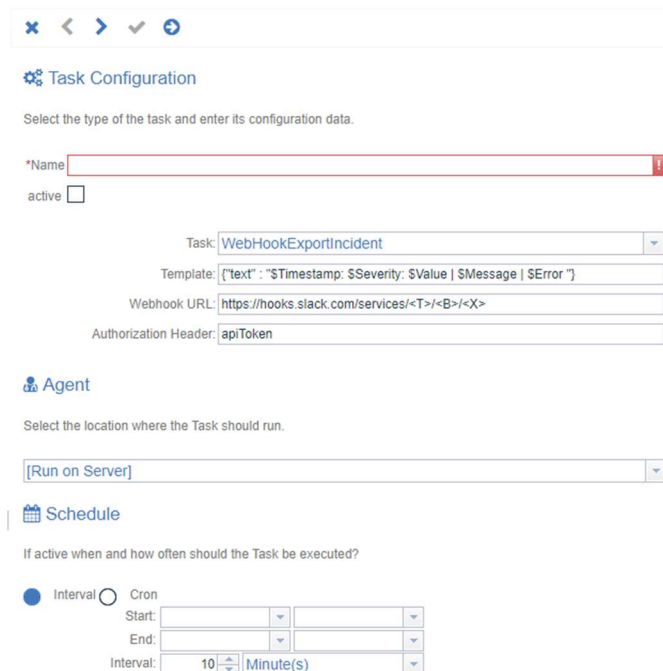
3.3 Setting up the WebHookExportIncident Task

Example for MS Teams:

Open the ESM Console and switch to the configuration part. Select task in the left bar and click on the + next to WebHookExportIncident to create a new task:



The default task setup looks like this. The given parameters must be overwritten:



Task Configuration

Select the type of the task and enter its configuration data.

*Name

active ☐

Task:

Template:

Webhook URL:

Authorization Header:

Agent

Select the location where the Task should run.

Schedule

If active when and how often should the Task be executed?

☒ Interval ☐ Cron

Start:

End:

Interval:

First specify a name and then overwrite the template with “single line” message card JSON we created in Part 2. Also overwrite the Webhook URL with the URL you have created in Part 1. Make sure you don’t have any space in front or at the end of the entries. Do not make the task active, just save it with the button from above.

✕ ◀ ▶ ✓ ↻

Task Configuration

Select the type of the task and enter its configuration data.

*Name

active ☐

Task:

Template:

Webhook URL:

Authorization Header:

Agent

Select the location where the Task should run.

Schedule

If active when and how often should the Task be executed?

☒ Interval ☐ Cron

Start:

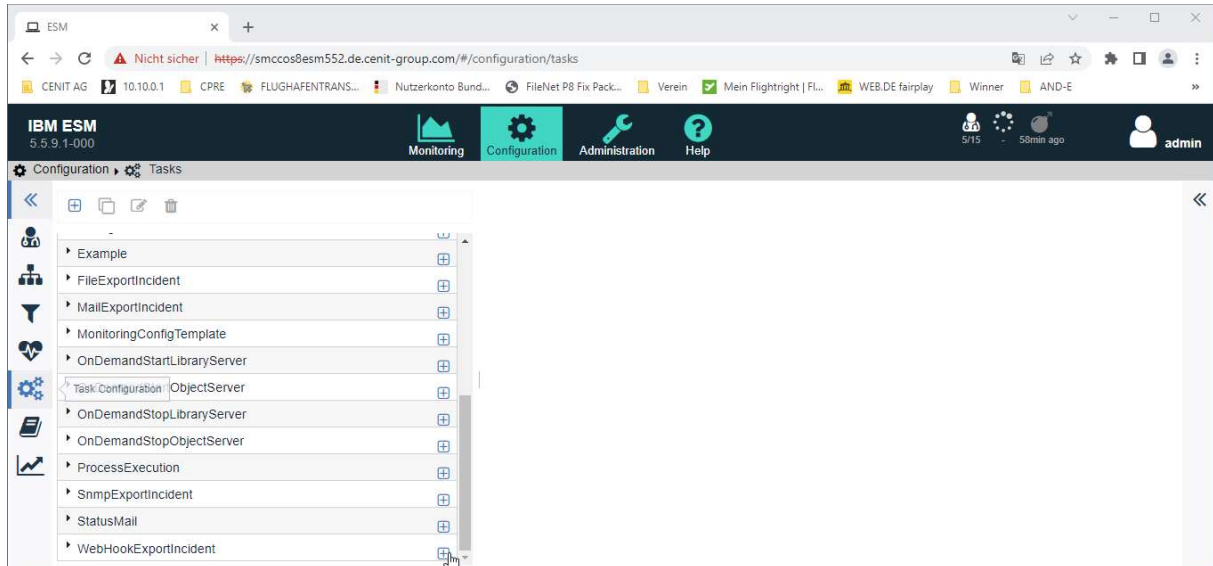
End:

Interval:

You can now setup the task as automation (see chapter 4).

Example for Slack:

Open the ESM Console and switch to the configuration part. Select task in the left bar and click on the + next to WebHookExportIncident to create a new task:



The default task setup looks like this. The given parameters must be overwritten:

Task Configuration

Select the type of the task and enter its configuration data.

*Name

active

☐

Task:

WebHookExportIncident

Template:

{"text" : "\$Timestamp: \$Severity: \$Value | \$Message | \$Error"}

Webhook URL:

https://hooks.slack.com/services/<T>//<X>

Authorization Header:

apiToken

Agent

Select the location where the Task should run.

[Run on Server]

Schedule

If active when and how often should the Task be executed?

☒ Interval ☐ Cron

Start:

End:

Interval:

10

Minute(s)

First specify a name and then overwrite the template with “single line” JSON we created in Part 2. Also overwrite the Webhook URL with the URL you have created in Part 1. Make sure you don’t have any space in front or at the end of the entries. Do not make the task active, just save it with the button from above. The “Authorization Header” might not be needed and can be removed.

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Task Configuration

Select the type of the task and enter its configuration data.

*Name

active ☐

Task:

Template:

Webhook URL:

Authorization Header:

Agent

Select the location where the Task should run.

Schedule

If active when and how often should the Task be executed?

☒ Interval ☐ Cron

Start:

End:

Interval:

You can now setup the task as automation (see chapter 4).

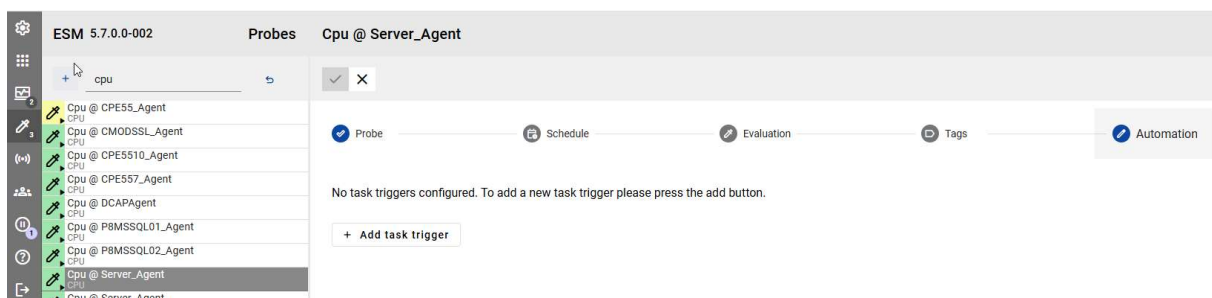
4. Automate forwarding

To automatically forward incidents by task there are two setup procedures available.

You can either set up the forwarding per Probe configuration individually, or you can set up global conditions in the task itself.

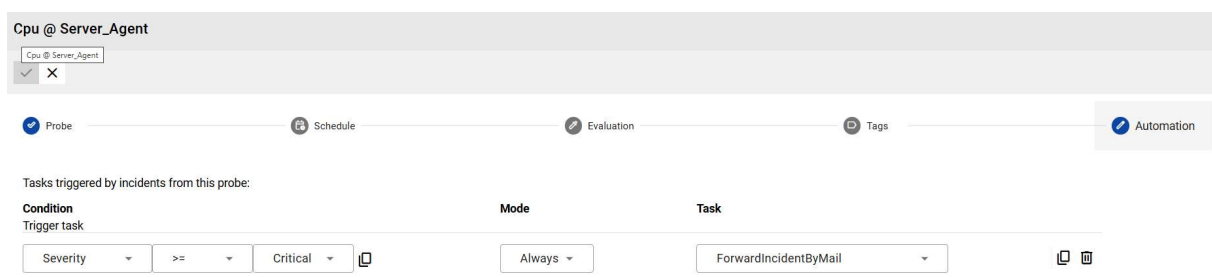
4.1 Automation setup in the Probe configuration - Simple CPU Probe example

In the Angular UI switch to the probes and select the related probe in the tree. Edit the configuration of the Probe so the probe editor opens. Switch to the last step (Automation) of the configuration.



Click on + Add task trigger to add a condition.

Add your condition as desired:



The Mode is either Always (whenever the condition matches) or First (only on the change of the severity – when the condition matches).

As Task select the task which you want to trigger. This example uses the Mail task created above, but you can use any of the tasks available.

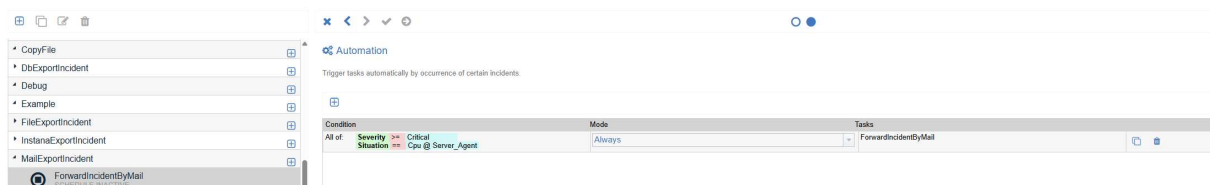
Save the setup.

4.2 Automation setup in the task configuration

This option is more global than the other setup, as the condition does not require the probe itself. E.g., it can be used to forward any Critical information as shown in this example.

The configuration for this is currently only available in the Configuration console via the task itself.

Open the task configuration as done during the task setup and switch to the second page (Automation).



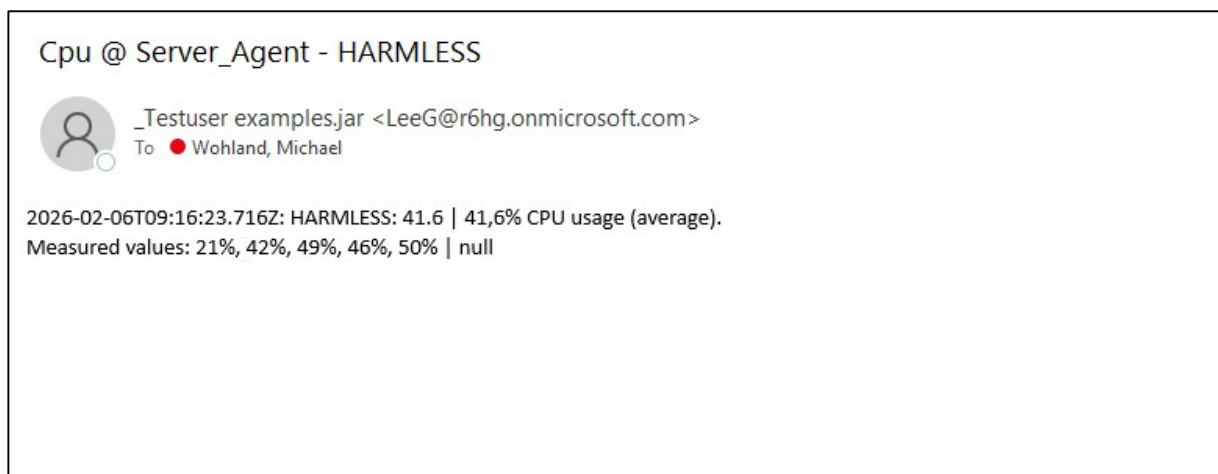
You can see the condition that was created in the Probe for the automation in here as well.

Click on the + to add an additional condition. For this more global setup the Situation is not required, and it can be removed once the condition is shown - Example:



Save the setup.

No matter which of the two options you select, you should now receive an email whenever the condition matches.



Contact Information

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